Metal Science and Heat Treatment, Vol. 55, Nos. 1 – 2, May, 2013 (Russian Original Nos. 1 – 2, January – February, 2013)

TITANIUM AND ZIRCONIUM ALLOYS

UDC 669.295.5'293'294'296

MECHANICAL AND ELECTROCHEMICAL CHARACTERISTICS OF THERMOMECHANICALLY TREATED SUPERELASTIC Ti – Nb – (Ta, Zr) ALLOYS

V. A. Sheremet'ev,[1] S. M. Dubinskii,[1, 2] Yu. S. Zhukova,[1] V. Brailovski,[2] M. I. Petrzhik,[1] S. D. Prokoshkin,[1] Yu. A. Pustov,[1] and M. R. Filonov[1]

Translated from *Metallovedenie i Termicheskaya Obrabotka Metallov*, No. 2, pp. 43 – 52, February, 2013.

Such parameters of superelasticity as elastic modulus, residual strain, phase yield point and mechanical hysteresis are determined by cyclic mechanical tests in a tension-unloading mode (2% maximum deformation, 10 cycles) for alloys of the Ti – Nb – Ta and Ti – Nb – Zr systems to be used as materials for medical implants. The testing is preceded by a thermomechanical treatment including multiple cold deformation and post-deformation annealing at various temperatures with water cooling. The stability of the parameters of superelasticity in subsequent aging at room temperature and repeated tests is studied. The annealing modes yielding an oxide film with maximum cohesion strength are determined.

Key words: titanium alloys, shape memory, thermomechanical treatment, mechanical cyclic tests, mechanical properties, oxide film, cohesion strength, electrochemical characteristics.

INTRODUCTION

The high-strength corrosion-resistant alloys based on titanium and cobalt and stainless steel traditionally used as materials for implants in medicine differ from bone tissue markedly; their modulus of elasticity is much higher than that of bone. This difference disturbs the mechanicobiological equilibrium in the organism. Appearance and growth of new cells of bone tissue is impossible in the absence of elastic deformations of the bone surrounding the much more rigid implant. This results in degradation of the bone near the implant and progressive looseness. Thus, the problem of biological compatibility of the material of implants is very urgent in the science of medical metals.

Shape memory alloys (SMA) possessing superelasticity (pseudoelasticity) are the closest to bone tissue in their mechanical behavior [1, 2]. The traditional alloys based on the TiNi intermetallic contain toxic nickel, which limits their use in medicine. Titanium nickelide can be replaced by SMA based on the Ti – Nb system, where a reversible martensitic transformation occurs too. These are alloys of the Ti – Nb – Ta and Ti – Nb – Zr system [3 – 7] which contain only biologically compatible components.

It has been shown in [8 – 11] that the functional properties of SMA can be controlled with the help of thermomechanical treatment (TMT) that involves cold or warm deformation (from moderate to severe) and post-deformation annealing (PDA). As a result of the TMT the alloy acquires a developed polygonized dislocation substructure (including a nanosubgrain one) or a nanocrystalline structure in the initial high-temperature phase, i.e., austenite [11, 12]. The formation of a nanocrystalline structure including a combination with a nanosubgrain substructure provides an ultimately high set of functional properties, both "static" and "dynamic" (fatigue), in Ti – Ni shape memory alloys [12 – 14]. The experience of control of the structure, substructure and functional properties of Ti – Ni alloys by TMT has been applied to SMA of the Ti – Nb – Ta and Ti – Nb – Zr systems. Some experiments with Ti – Nb – Ta and Ti – Nb – Zr alloys have been devoted to the special features of their superelastic behavior [5 – 7]. Fatigue tests under the conditions of implementation of superelasticity have made it possible to deter-

[1] National University of Science and Technology "MISiS," Moscow, Russia (e-mail: sheremetev88@yandex.ru).
[2] Ecole de Technologie Superieure, Montreal, Canada.

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mine the range of TMT regimes forming a nanosubgrain structure and providing the highest endurance and the best superelastic behavior in Ti – Nb – Zr alloys [15, 16]. Specifically, a gradual decrease in the modulus of elasticity and residual strain has been detected upon growth in the number of TMT cycles. At the same time, it has been shown in [17] for alloy Ti – 24% Nb – 4% Zr – 7.95% Sn that the value of the modulus of elasticity recovers after a six-day hold after cyclic tests. This makes the study of the stability of the modulus of elasticity and other characteristics of loading-unloading curves for Ti – Nb – Ta and Ti – Nb – Zr alloys very important.

It is also important to create a "friendly" surface on the implant for favorable interaction with bone tissue. Therefore, it is interesting to study the cohesion properties of the surface film forming in the final stage of the TMT, i.e., in annealing of SMA based on the Ti – Nb system.

During service, the implant is in contact with the corrosion-active biological environment of the human body, which makes it important to study the electrochemical behavior of Ti – Nb – Ta and Ti – Nb – Zr alloys in physiological solutions simulating such environments [18].

The aim of the present work was to determine the stability of the characteristics of superelasticity of Ti – Nb – Ta and Ti – Nb – Zr alloys under the conditions of mechanical cycling and subsequent aging, to estimate the cohesion strength of the surface oxide film after various modes of TMT, and to determine the characteristics of the electrochemical behavior of the alloys in environments imitating human body liquids.

METHODS OF STUDY

We studied alloys of the Ti – Nb – Zr and Ti – Nb – Ta systems.

Alloys TNZ1 and TNZ2 of the Ti – Nb – Zr system (with compositions[3] Ti – 21.8% Nb – 6% Zr and Ti – 20.8% Nb – 5.5% Zr respectively) were obtained by vacuum induction melting followed by hot pressing at 950 – 900°C. After the pressing, the ingots had the form of cylinders 50 mm in diameter and 660 mm long.

Alloy TNT1 of the Ti – Nb – Ta system (Ti – 19.7% Nb – 5.8% Ta) was melted in a vacuum induction furnace with a consumable electrode through an Nb + Ta alloy addition. The ingot with a mass of 7 kg and diameter 80 mm was subjected to hot forging and turning to a diameter of 50 mm. Alloy TNT2 (Ti – 19.5% Nb – 6.5% Ta) was obtained by vacuum electric arc melting in the form of ingots with a lenticular shape 50 mm in diameter with a thickness of 15 mm in the center.

The ingots were cut into billets by an electric spark and by an abrasive wheel. The billets with a size of 5 × 7 × 150 mm (TNT1) and 4 × 4 × 150 mm (TNZ1) were subjected to a multistage TMT that included rotation forging and cold drawing (CD) with intermediate annealing. The drawing gave wires with diameter ∅ 0.53 mm (TNZ1) and ∅ 0.7 and 1.35 mm (TNT1), which were subjected to a finishing annealing and used for mechanical tests. The strain $e = \ln F_0/F_f$ accumulated after the finishing annealing of these alloys was 1.38, 1.07 and 0.56 respectively (here F_0 and F_f are the initial and final areas of the cross sections of the specimens).

TABLE 1. Modes of Thermomechanical Treatment of Alloys of Ti – Nb – Ta (TNT) and Ti – Nb – Zr (TNZ) Systems

Alloy	e	t_{pda}, °C
TNT1	0.56 (CD)	550, 600 (1 h) and 600 (0.5 h)
TNT2	0	500, 550, 600, 650, 700, 750, 800 (1 h)
TNZ1	1.07; 1.38 (CD)	550, 600 (1 h) and 600 (0.5 h)
TNZ2	0.75 (CR)	500, 550, 600, 650 (1 h)

Notations: CD) cold drawing; CR) cold rolling; e) strain ratio ($e = \ln h_1/h_0$ in CR; $e = \ln F_0/F_f$ in CD), t_{pda}) temperature of post-deformation annealing for 1 h or 0.5 h (given in the parentheses).

To make flat specimens with a size of 2 × 10 × 150 mm (TNT2) and 2 × 8 × 20 – 100 mm (TNZ2) from billets, the alloys were subjected to TMT that included multistage cold rolling and PDA with cooling in water. The modes of the TMT and PDA of the alloys are presented in Table 1.

The specimens for mechanical testing at room temperature were subjected to PDA in a temperature range of 550 – 600°C. This regime was chosen because the range of the temperatures of post-deformation annealing most favorable for the effect of superelasticity in the alloys is a neighborhood of 600°C and corresponds to formation of a chiefly polygonized β-phase substructure with nanometric or submicron sizes of the structural components and a low content of ω-phase [15, 16, 19].

The tensile tests of specimens with a functional part 20 mm long fabricated from alloys TNT1 and TNT2 were performed in an "Instron 3360" testing machine in the mode "tensile deformation (with ε = 2%) – unloading" (10 cycles). Forty days after the test the same specimens were subjected to second testing in the same mode. The parameters of the obtained loading-unloading curves were determined by the method described in [20].

To study the oxide film on the surface of heat treated alloys by the method of measuring scratching, specimens of alloys TNT2 and TNZ2 were cut in the shape of a rectangle 8 × 15 mm in size. Deformed and hardened specimens of alloy TNZ2 were subjected to electrolytic polishing in a "STRUERS ElectroPol-5" device. The electrolyte was composed of 190 ml $HClO_4$ (with density 1.54 – 1.57 g/cm^3) and 800 ml $C_4H_5O_3$. Hardened specimens of alloy TNT2 were subjected to mechanical polishing. The final treatment of the laps was performed in the STRUERS device using a

[3] Here and below the content of elements is given in atomic percent.

Chempol abrasive polishing suspension on a black woolen cloth. The ready laps were cleaned before the heat treatment by isopropyl alcohol under ultrasonic irradiation. To form an oxide film of the surface of alloys TNT2 and TNZ2 the latter were annealed by the regimes presented in Table 1.

We used a "REVETEST" device for determining the adhesion/cohesion strength by measuring scratching. During sliding of a Rockwell-C diamond cone indenter (with tip radius 200 μm) moving at a specified speed and increasing load, we recorded the readings of several sensors, i.e., the loading force, the intensity of the acoustic emission (AE), the friction force (FF), and the friction coefficient (FC). Analyzing the form of the "property – load" curves and observing the traces of fracture of the coating under a light microscope we determined the minimum (critical) load (L_c) that has caused the fracture and the mechanism of the fracture (adhesion or cohesion). The hardness H measured by the method of scratching was computed as the ratio of the vertical load P to the area of the projection of the spherical contact between the specimen and the indenter by the formula

$$H = \frac{P}{\frac{\pi D}{4}(D - \sqrt{D^2 - b^2})}. \quad (1)$$

The scratch width b before breakage of the film, i.e., under the critical load, was taken to be the diameter of the circle. For the electrochemical study the specimens before the measurement were sanded (against a sandpaper with granularity from 160 to 1200). In order to remove the hardened surface layer with a thickness of 100 nm the specimens were etched with Ar^+ ions. Right before the electrochemical measurements the specimens were degreased with an alcoholic solution, washed in distilled water and treated in an "Ultrasonic Cleaner ST-405." The model physiological environments were biological solutions of the following compositions:

1) a Hank solution imitating the inorganic component of bone tissue and composed of (in g/liter) 8 NaCl, 0.4 KCl, $0.12Na_2HPO_4 \cdot 12H_2O$, 0.06 KH_2PO_4, $0.2MgSO_4 \cdot 7H_2O$, 0.35 $NaHCO_3$, and 0.14 $CaCl_2$ with an addition of H_2O (to 1 liter of the solution); pH = 7.4;

2) a solution of artificial saliva (in g/liter) composed of 0.4 KCl, 0.4 NaCl, 0.795 CaCl, 0.69 Na_2HPO_4, $0.005Na_2S \cdot 9H_2O$ + 1 g/liter urea with an addition of H_2O (to 1 liter of the solution); pH = 7;

3) an acidified Hank solution imitating a traumatized condition of bone tissue; pH = 5.

The electrochemical measurements were performed using an "IPC-Micro" electronic potentiometer equipped with a software for continuous automatic recording of parameters by the method of chronopotentiometry. We used a two-electrode cell with a divided electrode space, which was placed into an ultrathermostat with glycerin for keeping the specified temperature of the solution (37°C). The accuracy of the maintenance of the temperature was ± 1°C. A saturated chloride-silver electrode was used as a reference one.

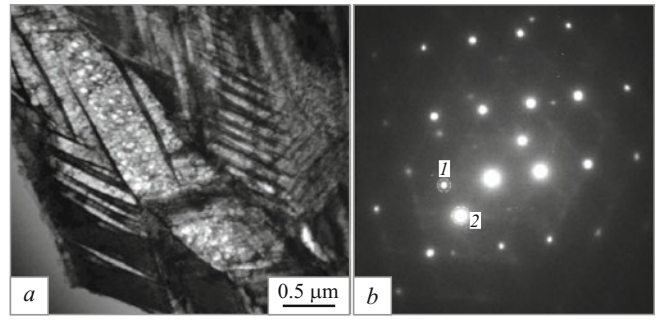

Fig. 1. Structure of alloy TNT1 against a light background (*a*) and a diffraction image (axis of zone $\langle 110 \rangle_\beta$) (*b*): *1* and *2*) reflections of the β-phase belonging to nets of different twinning orientations.

RESULTS AND DISCUSSION

Mechanical Tests

The metallic materials used for implants should imitate the mechanical behavior of bone tissue, i.e., possess a low modulus of elasticity and a property of superelasticity [1, 22]. By the data of [23] the modulus of elasticity of bone tissue is equal to 4 – 27 GPa; that of spongy tissue is equal to 1 – 11 GPa. For comparison, the modulus of elasticity of corrosion-resistant steels and cobalt alloys is about 200 GPa; that of titanium and titanium alloys (not superelastic ones) exceeds 100 GPa.

For mechanical tests we chose specimens of alloys TNT1 and TNZ1 subjected to a TNT that included CD + PDA at 600°C for 30 min, 600°C for 1 h, and 550°C for 1 h. According to the data of [15, 16] treatment by such regimes yields in a Ti – Nb – Zr alloy a nanosubgrain structure in the β-phase and the highest combination of functional properties. The electron microscope study showed that the PDA of alloy TNT1 at 600°C also promotes formation of a nanosubgrain structure in the β-phase (Fig. 1).

Figure 2 presents typical loading-unloading curves reflecting the change in their behavior in the course of cyclic tests of alloys TNZ1 and TNT1. The plotted loading-unloading curves (the stress-strain dependences) were used to determine the strain parameters (the modulus of elasticity E, the phase yield strength σ_{tr}, the mechanical hysteresis $\Delta\sigma$, and the residual strain ε_f after each cycle), as it is shown in Fig. 2*c*.

In the first loading-unloading cycle of the first series of tests of alloy TNT1 we observe imperfect superelasticity (an open "superelasticity loop"). In subsequent cyclic tests the superelastic behavior is improved progressively, and the residual strain is quite low already in the tenth cycle (Fig. 2*a*). The superelastic behavior of alloy TNZ1 in cyclic tests is improved more rapidly, and the residual strain is minimized al-

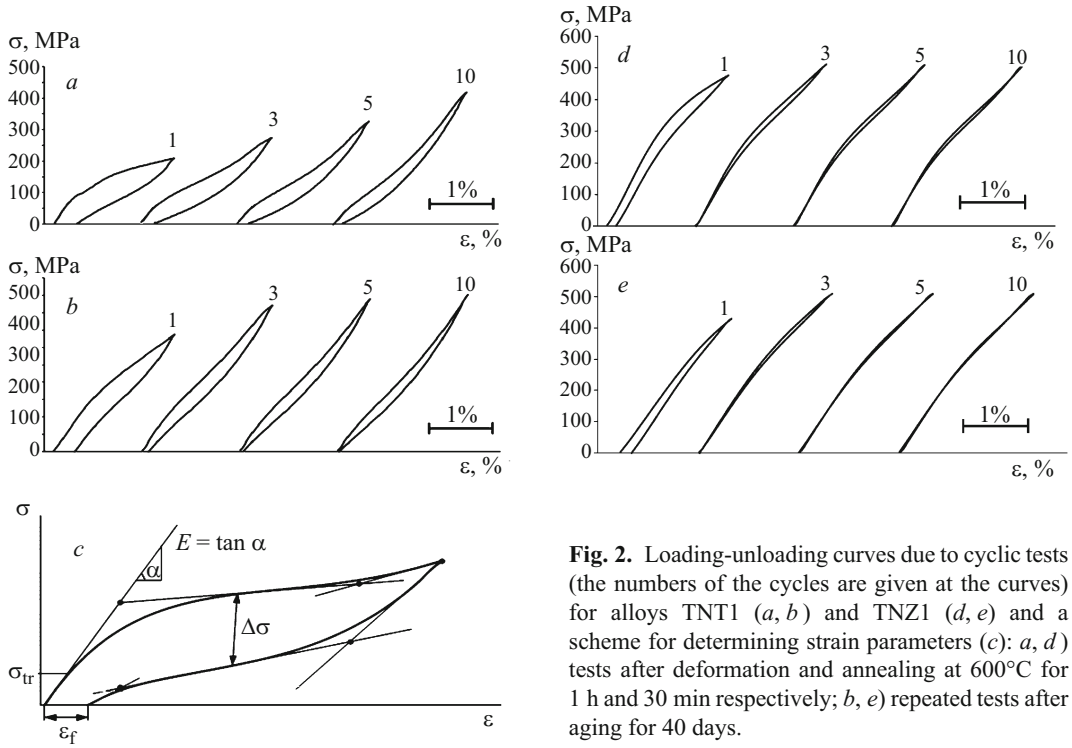

Fig. 2. Loading-unloading curves due to cyclic tests (the numbers of the cycles are given at the curves) for alloys TNT1 (*a*, *b*) and TNZ1 (*d*, *e*) and a scheme for determining strain parameters (*c*): *a*, *d*) tests after deformation and annealing at 600°C for 1 h and 30 min respectively; *b*, *e*) repeated tests after aging for 40 days.

ready after 3 – 5 cycles (Fig. 2*d*). Repeated tests of the alloys after aging for 40 days showed that in the fist cycle, just like in the first series of tests, the superelastic behavior is imperfect; there appears a noticeable residual strain (Fig. 2*b* and *e*). In the course of subsequent cyclic tests the picture is the same as in the first series of tests; the superelastic behavior is improved rapidly (Fig. 2*b* and *e*).

Figure 3 presents typical parameters of loading-unloading curves of the alloys after the same modes of TMT and the variation of the parameters in cyclic tests before and after aging. The behavior of two parameters, i.e., the modulus of elasticity and the residual strain, in time and due to thermocycling presents the greatest interest. It is desirable to obtain a stably low modulus of elasticity and a minimum residual strain.

It can be seen from Fig. 3*a* and *e* that the modulus of elasticity of the thermomechanically treated alloys is low, i.e., 45 – 50 GPa in the first loading-unloading cycle. In the process of cyclic testing it decreases tending to the modulus of elasticity of a dense bone tissue (25 – 30 GPa). In the Ti – Nb – Ta alloy its value is lower than in the Ti – Nb – Zr alloy. Aging for 40 days after the first series of tests does not cause substantial changes in the moduli of elasticity in both alloys (in contrast to the Ti – Nb – Zr – Sn alloy in [17], where the modulus of elasticity recovered by more than a half after 6 days). The attained value of the modulus of elasticity remains stable after repeated tests too (Fig. 3*a* and *e*).

The residual strain ε_f in the first cycle is about 0.25% for alloy TNT1 and about 0.15% for alloy TNZ1 (Fig. 3*b* and *f*). Forty-days aging after cyclic testing causes "recovery" of ε_f in the first cycle, but it lowers rapidly in subsequent cycling, i.e., the variation of ε_f is as if "reproduced" (Fig. 3*b* and *f*).

The phase yield strength σ_{tr} and the mechanical hysteresis $\Delta\sigma$ behave like ε_f, i.e., they decrease in the cyclic tests of the first series, "recover" after 40-days aging, and decrease again in subsequent cycling (Fig. 3*c*, *d*, *g* and *h*).

It should be noted that the phase yield strength of alloy TNZ1 is much higher and the mechanical hysteresis is much lower than in alloy TNT1. It should be borne in mind that the differences in the characteristics of the superelastic behavior of the Ti – Nb – Zr and Ti – Nb – Ta alloys (the values of the modulus of elasticity, of the phase yield strength, of the mechanical hysteresis, and of the residual strain and their stability in repeated testing) may be caused not only by the differences in their chemical compositions. It is quite probable that these differences are a consequence of different conditions of the tests, i.e., the difference between the test temperatures and point M_s of the Ti – Nb – Zr and Ti – Nb – Ta alloys is most likely different; in the former alloy it is higher.

The change in the mode of PDA (550°C for 1 h – 600°C for 30 min and 1 h) does not produce a noticeable effect on the characteristics of the superelastic behavior of alloy TNZ1. It is obvious that the intermediate short-term annealing at a moderate temperature during the drawing has stabilized the structure.

Thus, cyclic tests of thermomechanically treated nanostructured shape memory Ti – Nb – Zr and Ti – Nb – Ta alloys decreases the modulus of elasticity and brings it closer to that of a dense bone tissue. The attained value of the modulus of elasticity remains stable after 40-days aging and repeated cycling. The other parameters of the loading – un-

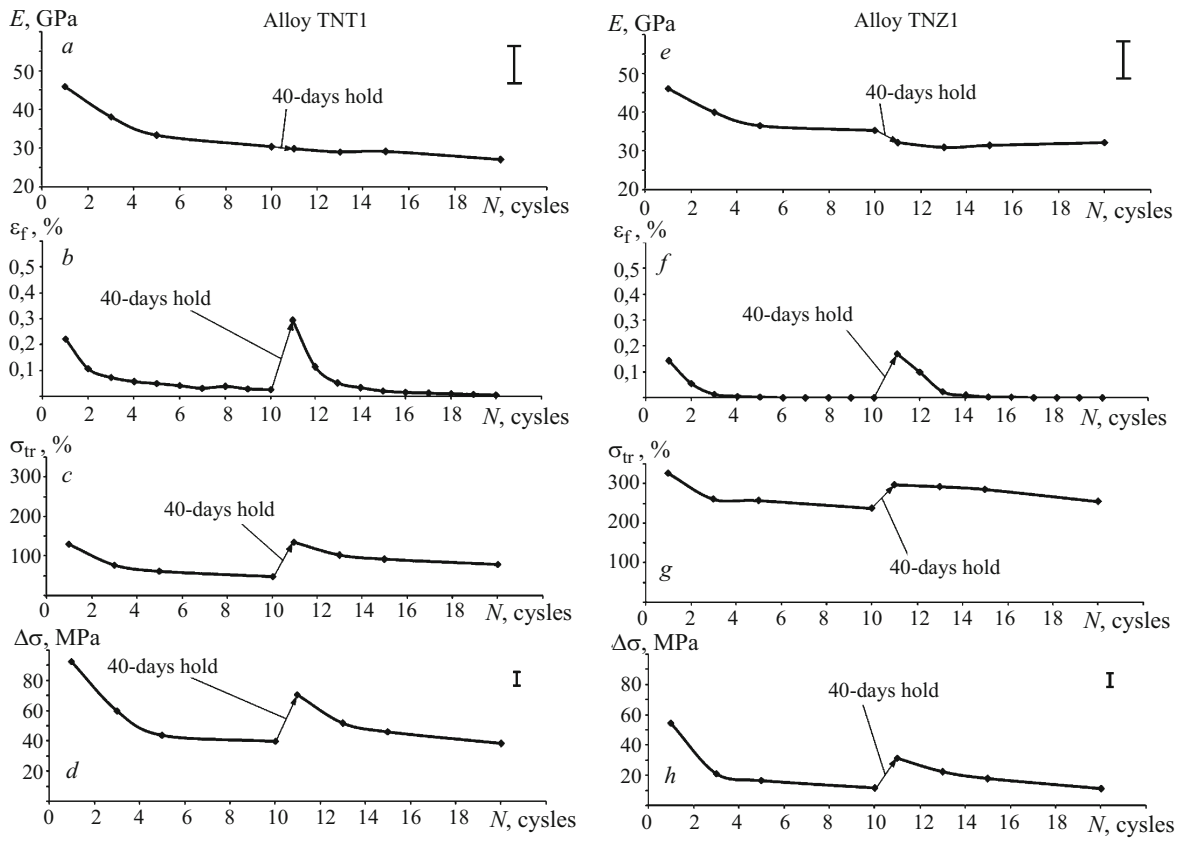

Fig. 3. Variation of the parameters of loading-unloading curves of thermomechanically treated alloys TNT1 and TNZ1 after annealing at 600°C for 30 min in the first series of cyclic tests and after aging for 40 days in repeated tests: *a*, *e*) modulus of elasticity *E*; *b*, *f*) residual strain; *c*, *g*) phase yield strength σ_{tr}; *d*, *h*) mechanical hysteresis $\Delta\sigma$.

loading curves stabilize only due to cyclic tests and do not exhibit time stability.

Measuring Scratching

The presence of an oxide film on the surface of a titanium alloy and the properties of the film affect directly its biological compatibility. To form an oxide film on the surface we annealed alloy TNT2 after preparing the surface of the specimens not subjected to preliminary deformation. The specimens of alloy TNZ2 were annealed after cold deformation with $e = 0.75$.

As a result of the annealing the surface of the alloy acquired metallic luster due to formation of an oxide film. The color of the film on the alloys annealed at 500 – 750°C varied from golden yellow to blue violet; after annealing at 800°C the surface of alloy TNT2 acquired a gray color.

Microscopic observations in the process of measuring scratching allowed us to determine the load at which the oxide film fractured and the mechanism of the fracture. Micrographs like that presented in Fig. 4*a* can be used for determining the "oxide-film – metallic matrix" transition, which is detectable from the appearance of metallic luster and from the change in the texture on the bottom and sides of the scratch when the load on the indenter is increased. The film had not been detached from any of the specimens, which allows us to classify the mechanism of its fracture as a cohesion one.

Microscope observations of a scratch correlate with the readings of the sensors detecting the friction force FF, the friction coefficient FC, and the acoustic emission AE. An example is given in Fig. 4*b* for the case of annealing at 500°C. All the curves except for those obtained at 800°C exhibit a high signal of acoustic emission in the studied load range, which reflects continuous fracture of the surface film. The signal of acoustic emission for the specimen of alloy TNT2 after a 1-h hold at 800°C is weak, which means that the surface film is loose and does not resist the fracture caused by the indenter.

The friction coefficient of the "diamond – oxide film" pair does not exceed 0.2 and increases markedly (to 0.9 – 0.95) upon contact between the indenter and the metallic matrix. This transition also allows us to judge on the critical load that has caused fracture of the oxide film.

We determined the critical load at which the film broke with allowance for the variation of the friction coefficient and of the acoustic emission and with allowance for the results of microscope observations under the assumption that this load was a characteristic of the cohesion strength. We also computed the hardness of the metallic matrix using the

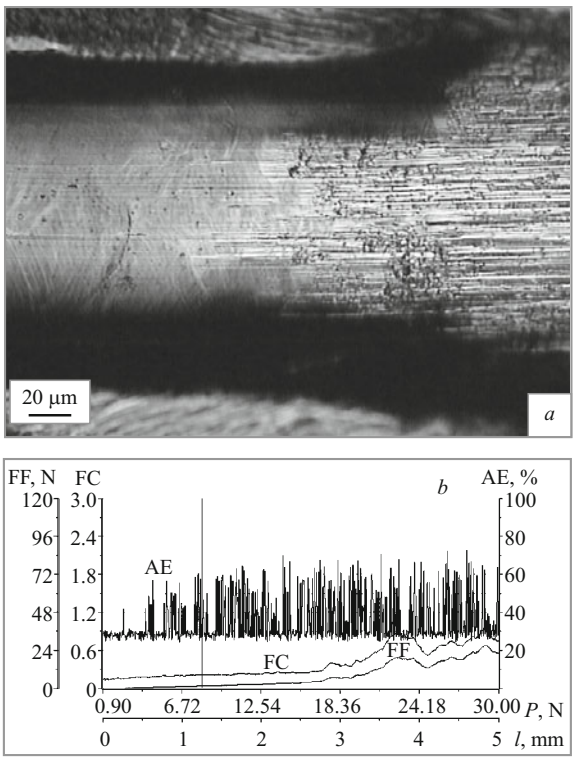

Fig. 4. Microstructure of a part of a scratch in the region of breakage of oxide film on the surface of alloy NTN_2 after annealing at 500°C (*a*) and variation of sensor signals (acoustic emission AE, friction force FF, friction coefficient FC) in the process of scratching of alloy TNT2 after annealing at 500°C under a load growing from 0.9 to 30 N (*b*): *P*) load on the indenter; *l*) indenter path.

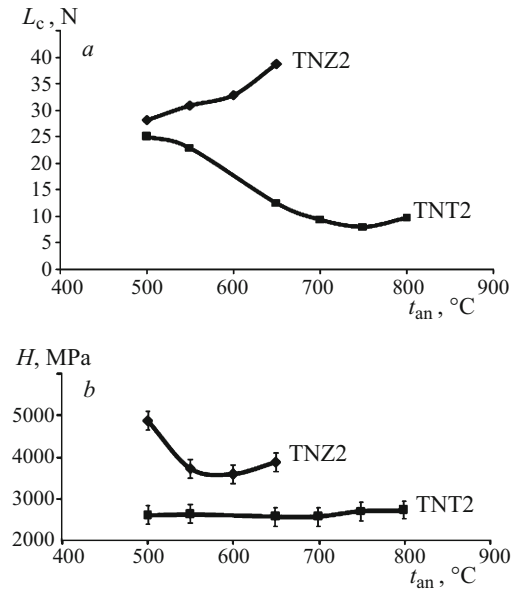

Fig. 5. Dependence of the critical load L_c (*a*) that causes breakage of oxide film on the surface of alloys TNT2 and TNZ2 and of the hardness *H* determined by the method of scratching (*b*) on the annealing temperature t_{an}.

results of measuring scratching [21] with allowance for the width of the scratch formed upon application of critical load to an indenter with a known tip radius.

The dependence of the cohesion strength on the annealing temperature is presented in Fig. 5*a*, from which it follows that the level of the cohesion strength of alloy TNZ2 is higher than that of alloy TNT2 and grows upon increase in the temperature. The cohesion strength of alloy TNT2 remains virtually invariable for the annealing temperature of 500 – 550°C and decreases when the temperature is increased to higher values. The difference in the characteristics of the cohesion strength of the alloys and its variation in opposite directions upon growth in the annealing temperature is explainable by different methods of preparation of the surface of the specimens (mechanical one for TNT2 and electrolytic one (polishing) for TNZ2) and kinds of formation of the structure of the substrate (heat treatment for TNT2 and thermomechanical treatment for TNZ2).

The hardness (Fig. 5*b*) of alloy TNZ2 determined by the method of measuring scratching is higher than that of alloy TNT2. For the TNT2 alloy it is independent of the annealing temperature. Alloy TNZ2 has the highest hardness after PDA at 500°C. The growth in the hardness at this annealing temperature is explainable by hardening of Ti – Nb – Zr alloys due to formation of ω-phase during the hold at this temperature [19]. When the annealing temperature is increased, the ω-phase dissolves in the equilibrium β-phase. At higher annealing temperatures the hardness stabilizes. In the Ti – Nb alloys with tantalum formation of ω-phase in heating has not been detected [24]. It should also be taken into account that alloy TNZ2, in contrast to TNT2, has been subjected to a deformation treatment, which raises the hardness.

Electrochemical Behavior

The behavior of the free electrode potential E_{fr} during exposure in model solutions is an important tool for studying the susceptibility to passivation and the laws of formation of protective films on the surface of alloys.

Figure 6 compares chronopotentiograms of alloys TNT1 and TNZ1 in model biological solutions *1*, *2* and *3* at 37°C.

To determine the mechanism of formation of passivating films and the kinetic laws of their growth on the surface of the alloys during exposure in the model solutions the chronopotentiograms were processed with allowance for the proportional dependence of the thickness of the film on the potential. For every moment of time τ and the corresponding potential E_τ we computed a normalized change in the potential, i.e.,

$$e_\tau = \frac{\Delta E_\tau}{\Delta E} = \frac{E_\tau - E_0}{E_\infty - E_0}. \quad (2)$$

Figure 7 gives an example of plotting of a normalized chronopotentiogram for alloy TNT1 in solution *1* at 37°C.

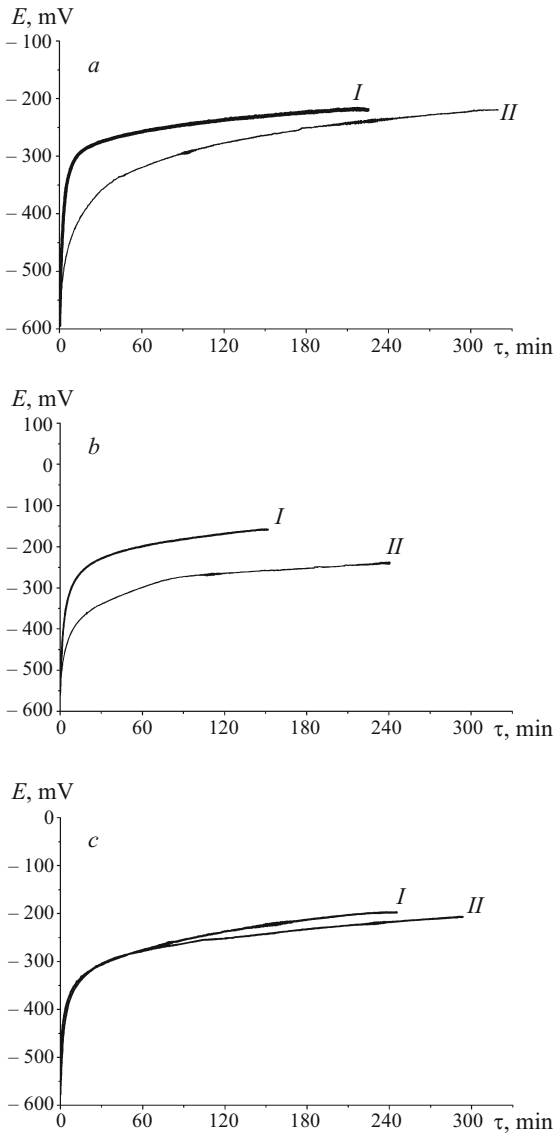

Fig. 6. Chronopotentiograms of alloys TNT1 (curve *I*) and TNZ1 (curve *II*) in model biological solutions *1* (*a*), *2* (*b*) and *3* (*c*) at 37°C.

Analysis of the chronopotentiograms obtained shows that all the normalized chronopotentiograms can be approximated with a high degree of reliability (with correlation coefficient $R^2 > 0.99$, Table 2) by two analytical logarithmic dependences describing the variation of e_τ in two time intervals (zone *A* and zone *B*). The time boundary (τ_x) between zones *A* and *B* was chosen visually and from the highest value of the correlation coefficient in the approximated zones *A* and *B*. These dependences are described by the following equations in zones *A* and *B* respectively:

$$e = a + k \ln(\tau + c), \quad (3)$$

$$e = \frac{1}{a - k \ln(\tau + c)}, \quad (4)$$

where *a*, *k*, and *c* are constants.

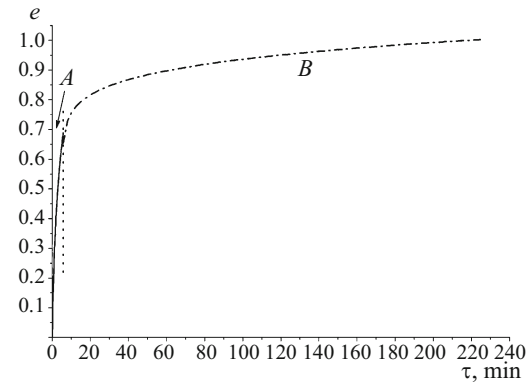

Fig. 7. Normalized chronopotentiogram of alloy TNT1 in solution *1* at 37°C (reference electrode: saturated chloride-silver electrode): *A*) zone of existence of dependence (3); *B*) zone of existence of dependence (4).

The kinetic laws corresponding to Eqs. (3) and (4) have been suggested and substantiated by the authors of [25] and [26] for describing the growth of thin oxide films (finer than 5 mm) at low temperatures including the case of anode oxidation of metals. Within their concept very thin films should form according to a direct logarithmic dependence with subsequent transition to an inverse logarithmic dependence. The rate of formation of the thinnest layers is determined by the speed of motion of electrons with the help of tunnel effect. At a low temperature the diffusion of ions through the film is hindered, while the motion of electrons provides high conductivity of the oxide film. Metal cations form on the internal surface of the metal/oxide interface, and oxygen anions (or anions of another oxidant) form on the external surface, which gives rise to a strong electric field inside the oxide film, which is capable to "tear" a metal ion and to transfer it through the oxide layer. The rate of growth of the films is determined just by this very slow process. Constants *k* in Eqs. (3) and (4) are in fact the constants of the rate of growth of oxide passivating layers in the appropriate time zones.

The results of an approximation of normalized chronopotentiograms are presented in Table 2. It can be seen that alloy TNT1 manifests the highest rate of formation of passivating layers (the highest values of constant *k* in zone *A*). The time of the change of the dominant mechanism of growth of the film for the TNT1 Ti – Nb – Ta alloy is lower than that for alloy TNZ1.

CONCLUSIONS

1. Cyclic mechanical tests of Ti – Nb – Zr and Ti – Nb – Ta alloys with a polygonized (nanosubgrain) structure in a "tension – unloading" mode produce the phenomenon of imperfect superelasticity in the first cycle; in subsequent cycling this property is advanced, which occurs more rapidly in the Ti – Nb – Zr alloy.

TABLE 2. Results of Approximation of Normalized Chronopotentiograms of Alloys TNT1 and TNZ1 in Model Biological Solutions

Alloy	Solution	τ_x, min	Zone	a	k	c	R^2
TNT1	I	6	*A*	0.06 ± 0.01	0.33 ± 0.00	0.76 ± 0.02	0.998
			B	1.45 ± 0.00	0.08 ± 0.00	– 5.45 ± 0.01	0.999
	II	8	*A*	0.14 ± 0.00	0.24 ± 0.00	0.48 ± 0.01	0.998
			B	1.59 ± 0.00	0.12 ± 0.00	– 6.93 ± 0.01	0.999
	III	12	*A*	0.16 ± 0.00	0.19 ± 0.00	0.34 ± 0.01	0.997
			B	1.78 ± 0.00	0.14 ± 0.00	– 9.17 ± 0.02	0.999
TNZ1	I	43	*A*	– 0.04 ± 0.00	0.19 ± 0.00	1.15 ± 0.00	1.000
			B	1.94 ± 0.00	0.17 ± 0.00	– 29.36 ± 0.06	0.998
	II	55	*A*	0.07 ± 0.00	0.18 ± 0.00	0.558 ± 0.00	0.999
			B	1.42 ± 0.00	0.08 ± 0.00	– 48.16 ± 0.09	0.991
	III	25	*A*	0.16 ± 0.00	0.15 ± 0.00	0.26 ± 0.00	0.999
			B	2.16 ± 0.00	0.22 ± 0.00	– 11.77 ± 0.06	0.999

Notations: τ_x) time of change of the dominant mechanism of growth of oxide film: k) constant in Eqs. (3) and (4) characterizing the rate of growth of passivating oxide films in the corresponding time zones; a, c) constants in Eqs. (3) and (4); R^2) correlation coefficient.

Note. In all the tests the solutions have a temperature of 37°C.

2. The modulus of elasticity of mechanically treated Ti – Nb – Zr and Ti – Nb – Ta alloys is low and decreases in the process of cyclic testing ($N = 10$ cycles) approaching the modulus of elasticity of bone tissue (25 – 30 GPa); in the Ti – Nb – Ta alloy it is lower than in the Ti – Nb – Zr alloy. Thus, the thermomechanically treated Ti – Nb – Zr and Ti – Nb – Ta alloys in a nanostructured or ultrafine-grained condition possess high biomechanical compatibility to bone tissue.

3. The value of the modulus of elasticity attained after the tenth test cycle changes little in subsequent aging for 40 days and in repeated cyclic tests.

4. The residual strain, the phase yield strength, and the mechanical hysteresis decrease in the process of superelastic cyclic tests and then recover after 40-days aging; in repeated cycling these characteristics decrease again.

5. The oxide film formed during annealing of alloy 74% Ti – 19.5% Nb – 6.5% Ta and post-deformation annealing of alloy 73.7% Ti – 20.8% Nb – 5.5% Zr fractures by the cohesion mechanism. The maximum cohesion strength is observed in the Ti – Nb – Ta alloy after annealing at 500 – 550°C and in the Ti – Nb – Zr alloy after annealing at 650°C. After annealing at 800°C the oxide film on the alloys becomes loose and weak. The hardness determined by the method of measuring scratching is higher in the Ti – Nb – Zr alloy than in the Ti – Nb – Ta alloy; in the Ti – Nb – Ta alloy it is independent of the annealing temperature.

6. The method of chronopotentiometry has been used to study the kinetic features of the establishment of steady electrode potentials in the Ti – Nb – Zr and Ti – Nb – Ta alloys during exposure in physiological solutions. A mechanism of formation of protective passive films on the surface of the alloys has been suggested. The process of formation of protective films may be described by two logarithmic-type equations; according to the mechanism of formation and growth they resemble thin oxide films formed at low temperatures.

The work has been performed with financial support of the Ministry of Education and Science of the Russian Federation (State Task No. 7.2534.2011), of the Natural Science and Engineering Research Council of Canada, and of the Program for Creation and Development of the NITU "MISiS."

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